

Manufacturing Standard Operating Procedure (SOP)

Document Control

SOP Title

Clear and specific description of the procedure

SOP ID

Internal document reference number

Version

e.g. 1.0

Status

Draft / Approved / Released

Effective Date

DD Month YYYY

Supersedes

Previous SOP ID and version (if applicable)

Department / Area

Production, Maintenance, Site, Project, or Business Unit

Applicable Locations

Global / Regional / Site-specific

Prepared By

Name, Role

Reviewed By

Name, Role (Quality, EHS, Engineering)

Approved By

Name, Role (Management)

1. Purpose

The purpose of this Standard Operating Procedure is to define a clear and standardized method for performing the described activity. This SOP ensures consistent quality, safe execution, and compliance with internal and external requirements.

2. Scope and Applicability

This SOP applies to all personnel involved in executing, supervising, or verifying the described process. It is valid for all locations and operations listed above.

Activities outside the defined scope, as well as deviations from this procedure, must be formally approved and documented.

3. Responsibilities

Clear responsibilities ensure correct execution and accountability.

- **Operator**
Executes the procedure as described and reports deviations or abnormalities.
- **Supervisor**
Ensures the procedure is followed and required resources are available.
- **Quality / EHS**
Verifies compliance with quality and safety requirements and supports audits.
- **Engineering / Technical Authority**
Ensures technical correctness of parameters, tools, and methods.
- **Management**
Approves this SOP and ensures organizational compliance.

4. Definitions and Abbreviations

To avoid misinterpretation, the following definitions apply:

- **SOP** – Standard Operating Procedure
- **PPE** – Personal Protective Equipment
- **OOS** – Out of Specification

Additional terms specific to this procedure shall be defined as required.

5. References and Applicable Standards

This SOP is aligned with relevant standards and regulatory requirements, including where applicable:

- ISO 9001 – Quality Management Systems
- ISO 45001 – Occupational Health and Safety
- ISO 14001 – Environmental Management
- ISO 31000 – Risk Management
- Industry-specific regulations such as GMP or FDA requirements

Only the latest approved versions of referenced documents shall be used.

6. Safety and Risk Considerations

Safety is an integral part of this procedure.

Potential hazards associated with the activity must be identified and controlled. All mandatory safety measures and PPE requirements defined in this section shall be followed at all times.

 Failure to follow safety instructions may result in injury, equipment damage, or regulatory violations.

7. Materials, Tools, and Equipment

Procedures should clearly specify required materials, tools, and equipment. Only approved, suitable, and calibrated items should be used to ensure reliable and repeatable results.

8. Pre-Execution Requirements

Before execution, ensure that:

- The correct and approved SOP version is available
- Personnel are trained and authorized
- Equipment is released for use
- The work environment is safe and prepared

9. Procedure

This section describes the standardized execution of the process and its critical control points.

Step-by-Step Instructions

1. Prepare the workstation according to specification.
2. Set and verify all relevant parameters within defined tolerances.
3. Execute the operation in the defined sequence.
4. Perform required checks and confirm results.

Where necessary, photos, diagrams, or flowcharts shall be used to support execution.

10. Quality Checks and Acceptance Criteria

Quality checks ensure that outputs meet defined requirements.

- Inspection points and methods shall be clearly defined
- Acceptance criteria must be measurable and unambiguous
- Results shall be documented as required

11. Deviation and Escalation Procedure

If requirements cannot be met:

- Stop the process if safety or quality is affected
- Inform the defined responsible role immediately
- Document the deviation according to internal procedures
- Resume work only after approval or corrective action

Unauthorized deviations are not permitted.

12. Documentation and Records

All records generated during execution must be:

- Complete and legible
- Traceable to this SOP
- Stored according to internal retention and regulatory requirements

13. Training Requirements

Personnel performing this procedure must be trained and qualified.

Training records shall be maintained and made available for internal or external audits.

14. Revision History

Version | Date | Description | Approved By